

Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: **C. A. Gunter and D. S. Scott**, “**Semantic Domains**,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([../papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

Progress (as of r0009, 2026-0806)

#	Quantity	Done	Remaining	Of
1	Definitions to define	6	7	≈13
2	Numbered results complete	2 (Lem 4, Thm 7)	26	28
3	Mathlib foundations reused	12	—	12
4	sorry in the development	—	0	—

Row 2 counts only the paper’s 30 **numbered** results (Theorems / Lemmas / Proposition 1–30) and so understates what is proved. The development contains 55 theorems, 15 definitions and 12 instances; of the theorems, these **eight are claims the paper makes in prose** rather than as numbered results, and all eight are formally verified:

#	Paper claim	Where	Lean
1	“the compact elements [of $P \rightarrow N$] are just the finite subsets of N ”	p. 9	<code>isCompactElement_iff_finite</code>
2	“ $P \rightarrow N$... is a domain”	p. 9	<code>instance : Domain (Set X)</code>
3	“ $D \rightarrow E$ is a ... cpo”	Thm 7 proof	<code>instance : CompletePartialOrder (ScottHom α β)</code>
4	“... bounded complete ... whenever E is”	Thm 7 proof	<code>instance : BoundedComplete (ScottHom α β)</code>
5	“step(s) ... is continuous”	Thm 7 proof	<code>scottContinuous_stepFun</code>
6	“... and compact in the ordering on $D \rightarrow E$ ”	Thm 7 proof	<code>isCompactElement_step</code>
7	“they form a basis for $D \rightarrow E$ ”	Thm 7 proof	<code>instance : IsAlgebraic (ScottHom α β)</code>
8	every compact function is a <i>finite</i> join of step functions	Thm 7 proof, implicit	<code>exists_finite_isLUB_of_isCo</code>

Six of those eight are the body of **Theorem 7**, which is now **complete** — all four conjuncts of its conclusion (cpo, bounded complete, algebraic, countably based) are formally verified, as `ScottHom.isBoundedCompleteDomain_scottHom`.

It is proved under **weaker hypotheses than the paper states**: bounded completeness of D is never used. D need only be a domain. The function space is a cpo for any preordered D , algebraic when D and E are, bounded complete because E is, and countably based because D and E are.

The remaining ~47 theorems are supporting API: the $\ll$ calculus, the `compactBelow` machinery, the pointwise order and suprema on $D \rightarrow E$, and the step-function adjunction. The paper assumes or elides all of it.

4 of ≈13 definitions are formally verified and building, in 6 modules, 985 lines, 0 sorry, 0 warnings:

#	Round	Module	Contents
1	r0003	<code>ScottDomains/WayBelow.lean</code>	<code>way-below $\ll$ at [Preorder α], 7 theorems; <code>x $\ll$ x $\leftrightarrow$ IsCompactElement x</code> holds by <code>Iff.rfl</code></code>

#	Round	Module	Contents
2	r0004	ScottDomains/Domain.lean	IsAlgebraic, Domain (with the paper's countable-basis condition), BoundedComplete, 8 theorems, Domain Prop
3	r0005	ScottDomains/Powerset.lean	the paper's P N (p. 9): compacts of Set X are exactly the finite subsets, hence Domain (Set ℕ) — the nondegenerate witness
4	r0006–r0007	ScottDomains/ScottHom.lean	the continuous function space $D \rightarrow E$: ScottHom, the pointwise order, CompletePartialOrder, and BoundedComplete when E is — Theorem 7's first sentence in full
5	r0008	ScottDomains/StepFunction.lean	the single step function step k e: continuity (from k compact), the adjunction $\text{step } k \ e \leq f \leftrightarrow e \leq f \ k$, and compactness in $D \rightarrow E$ (from e compact)
6	r0009	ScottDomains/FunctionSpaceDomain.lean	Domain is algebraic — the paper's “they form a basis for $D \rightarrow E$ ”; the two halves of IsAlgebraic use disjoint hypotheses (directedness needs only E bounded complete; the lub needs only D, E algebraic)
7	r0010	ScottDomains/CompactFunction.lean	every compact function is a finite join of step functions — the finiteness half of the basis claim
8	r0011	ScottDomains/FunctionSpaceCountable.lean	Countable — D is countable, hence Domain (ScottHom α β) — Theorem 7 complete
9	r0012	ScottDomains/NormalSubposet.lean	the normal-subposet relation $\triangleleft$ and Lemma 4 , all four parts
10	r0012	ScottDomains/Projection.lean	embedding–projection pairs and projections; the paper's “an embedding is an injection, a projection is a surjection”

Next: finitary projections, which need $\text{im}(p)$ to carry a Domain structure as a subtype — a sub-cpo construction. Lemma 5 and Theorem 6 both quantify over them.

Work counts

- **Reuse from Mathlib — no work (12):** poset, directed, cpo, $\perp$, monotone, continuous, Fixed-Point Theorem (1) & operator, Schröder–Bernstein (2), algebraic lattice, product, λ -notation.
- **Generalize / adapt (4):** compact element (IsCompactElement — its *definition* is already stated at [PartialOrder α] and so applies to a dcpo verbatim; what is CompleteLattice-only is every lemma about it, so the dcpo API must be re-proved), sum, lift (WithBot/Sum), ideal completion (Order. Ideal).
- **Definitions to define — new (≈ 13 ; 4 done in r0003–r0004, 9 remaining):** way-below $\ll$ (**done** — ScottDomains/WayBelow.lean), algebraic cpo, **domain**, bounded-complete (**all three done** — ScottDomains/Domain.lean), embedding–projection pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and D^∞ .
- **Theorems to prove (28):** 28 of the paper's 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

Bottom line: ≈ 13 definitions to define + 28 results to prove, on top of 12 reused Mathlib foundations. After r0004: **9 definitions + 28 results** remain.

Lean column legend: ✓ reuse Mathlib (name given) · ~ partial (Mathlib has a related or lattice-only version — generalize) · ✕ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render $\rightarrow$ as $!$, drop fi ligatures, mangle math), so read them as a guide to *what* each result is. Notation: $\sqsubseteq$ order, $\sqcup$ directed sup, $\perp$ bottom, $\ll$ way-below, $K(D)$ compacts, $Fp(D)$ finitary projections.

§2 Recursive definitions of functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Def	—	poset (partially ordered set)	✓ <code>PartialOrder</code>
2.1	Def	—	directed subset M : every finite $u \sqsubseteq M$ has an upper bound in M	✓ <code>DirectedOn</code> / <code>IsDirected</code>
2.1	Def	—	cpo : poset in which every directed M has a lub $\sqcup M$	✓ <code>CompletePartialOrder</code> (chains: <code>OmegaCompletePartialOrder</code>)
2.1	Def	—	bottom $\perp$ (least element)	✓ <code>OrderBot</code>
2.1	Def	—	monotone function	✓ <code>Monotone</code> / <code>OrderHom</code>
2.1	Def	—	continuous : monotone, $f(\sqcup M) = \sqcup f(M)$ for directed M	✓ <code>ScottContinuous</code>
2.1	Thm	1	Fixed-Point Theorem : $f : D \rightarrow D$ continuous $\implies$ least fixed point $\sqcup_n f^n(\perp)$	✓ <code>OrderHom.lfp</code> (Kleene: <code>OmegaCompletePartialOrder</code>)
2.2	Thm	2	Schröder–Bernstein for sets	✓ <code>Function.Embedding.schroederBernstein</code>
2.3	Def	—	fixed-point operator (uniform)	✓ <code>OrderHom.lfp</code> / <code>LawfulFix</code>
2.3	Thm	3	The standard operator is the unique uniform fixed-point operator	✗ prove

§3 Effectively presented domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	compact element x : $x \sqsubseteq \sqcup M$ (dir.) $\implies x \sqsubseteq y$ some $y \in M$; $K(D)$	$\sim$ <code>IsCompactElement</code> — def is <code>[PartialOrder]</code> , so reusable on a <code>dcpo</code> ; its lemmas are all <code>CompleteLattice</code> -only
3.1	Def	—	way-below $\ll$ (approximation relation behind compactness)	✓ <code>ScottDomains.WayBelow</code> (r0003) — $x \ll x \leftrightarrow \text{IsCompactElement } x$ by <code>Iff.rfl</code>
3.1	Def	—	algebraic cpo: $x = \sqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)	✓ <code>ScottDomains.IsAlgebraic</code> (r0004)
3.1	Def	—	domain = algebraic cpo whose basis $K(D)$ is countable (the paper’s definition, p. 9 — the countability condition was missing from an earlier draft of this row)	✓ <code>ScottDomains.Domain</code> (r0004)
3.1	Def	—	bounded complete : $\perp$ + every bounded subset has a sup — a <i>separate</i> predicate; the paper composes them as “bounded complete domain” (Thm 7, Lem 10, Lem 13, Thm 14), which is the literature’s <i>Scott domain</i>	✓ <code>ScottDomains.BoundedComplete</code> (r0004); the compound is <code>[Domain α] [BoundedComplete α]</code>
3.1	Def	—	(countably based) algebraic lattice	✓ <code>IsCompactlyGenerated</code> (+ <code>CompleteLattice</code>)
3.1	Def	—	embedding–projection pair (g, f)	✓ <code>ScottHom.IsEmbeddingProjectionPair</code> (r0012)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	projection ; finitary projection p : $p \circ p = p \text{Eid}$, $\text{im}(p)$ a domain	$\sim \text{ScottHom.IsProjection}$ (r0012); the <i>finitary</i> condition needs $\text{im}(p)$ as a sub-cpo
3.1	Def	—	normal subposet / substructure	✓ $\text{ScottDomains.IsNormalIn}$, notation $\triangleleft$ (r0012)
3.1	Lem	4	$\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least	✓ proved (r0012), all four parts
3.1	Lem	5	p finitary projection $\implies$ compacts of $\text{im}(p)$ are $\{p(x) : x \in K(D)\}$	✗ prove
3.1	Thm	6	Isomorphism: normal substructures $\cong$ $\text{Fp}(D)$ (finitary projections)	✗ prove
3.1	Thm	7	D, E bounded-complete domains $\implies D \rightarrow$ E bounded-complete domain	✓ proved (r0006–r0011) — $\text{ScottHom.isBoundedCompleteDomain_scottHom}$; D bounded complete is not needed
3.2	Def	—	effective presentation $d : \mathbb{N} \rightarrow K(D)$; effectively presented domain	✗ define

§4 Operators and functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.1	Def	—	product $D \times E$	✓ Prod (order/cpo instances)
4.2	Def	—	Church's λ-notation (continuous abstraction)	✓ $\text{OrderHom} / \omega\text{CPO ContinuousHom}$
4.3	Def	—	smash product $D \otimes E$	✗ define
4.4	Def	—	sum $D + E$; lift $D\perp$	$\sim \text{Sum} / \text{WithBot, Part}$ (partial)
4.x	Lem	8	Currying/iso laws for function spaces over D, E, F	✗ prove
4.x	Lem	9	Product/function-space iso laws over D, E, F	✗ prove
4.5	Lem	10	D, E bounded complete $\implies \rightarrow, \times, \otimes, +, ()\perp$ bounded complete	✗ prove
4.5	Thm	11	Ideal completion of a countable pre-order is a domain (all domains so arise)	$\sim \text{Order.Ideal}$ exists $\rightarrow$ prove
4.5	Thm	12	Initiality of a continuous algebra satisfying axioms T	✗ prove
4.5	Lem	13	D bounded complete $\implies$ powerdomains $D], D[$ bounded complete	✗ prove
4.5	Thm	14	Equivalent characterizations of an (algebraic/BC) domain	✗ prove

§5 Powerdomains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
5.1	Def	—	powerdomain (non-deterministic outcomes)	✗ define
5.2	Def	—	Hoare (lower), Smyth (upper), Plotkin (convex) powerdomains	✗ define
5.3	—	—	Universal & closure properties (see Lem 13, 28, 30)	✗ prove

§6 Bifinite (SFP) domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.1	Def	—	Plotkin order / bifinite (SFP) domain: bilimit of finite pointed posets	✗ define
6.1	Prop	15	Every bounded-complete domain is bifinite	✗ prove
6.1	Thm	16	$D \text{ bifinite} \implies \text{Fp}(D)$ is an algebraic lattice	✗ prove
6.2	Lem	17	$D, E \text{ bifinite} \implies \rightarrow, \times, \otimes, +, ()^\perp$ bifinite (incl. function space)	✗ prove
6.2	Thm	18	If D and $D \rightarrow D$ are domains, then D is bifinite	✗ prove
6.2	Lem	19	closure $r: D \rightarrow D$ ($r \circ r = r \sqsupseteq \text{id}$) $\implies \text{im}(r)$ is a domain	✗ prove
6.2	Lem	20	$D \text{ domain} \implies \text{Fc}(D)$ (finitary closures) is a cpo	✗ prove

§7 Recursive definitions of domains (universal domain, D_∞)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Def	—	recursive domain equation; universal domain / D_∞ (inverse limit)	✗ define
7	Thm	21	F representable over cpo $U \implies$ a domain D with $D \cong F(D)$	✗ prove
7	Thm	22	Any countably-based algebraic lattice L : a closure $r : P(\mathbb{N}) \rightarrow L$	$\sim \text{IsCompactlyGenerated} \rightarrow$ prove
7	Lem	23	The function-space operator is representable over $P(\mathbb{N})$	✗ prove
7	Lem	24	$U \text{ cpo}; \times \text{ and } \rightarrow \text{ representable over } U \implies$ (setup for universality)	✗ prove
7	Thm	25	U non-trivial domain representing $\times, \rightarrow \implies$ U universal	✗ prove
7	Thm	26	Any signature $(s_1, \dots, s_n)$: combinators $F_1, \dots, F_n$ solving the equations	✗ prove
7	Thm	27	Any bounded-complete D : a projection of the universal domain onto D	✗ prove
7	Lem	28	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()^\perp, ()^\perp$ representable over U	✗ prove
7	Thm	29	$D \text{ bifinite} \implies D^+ \text{ bifinite}$; solving $D \cong D^+$	✗ prove
7	Lem	30	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()^\perp, ()^\perp$ p-representable over universal bifinite V	✗ prove

Tally (matched): ✓ reuse Mathlib $\approx$ **12** (poset, directed, cpo, $\perp$, monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product, λ -notation) · $\sim$ partial $\approx$ **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) · ✗ define / prove $\approx$ **44**, of which 4 are done ($\ll$ r0003; algebraic, domain, bounded-complete r0004) $\rightarrow$ **40 remaining** — the bifinite / powerdomain / D_∞ development and all 28 numbered results.

What is done, and what is next, is listed under **Progress** at the top of this file.